

fit@hcmus

# Software Testing

## CSC13003

Test case

# Test Case

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- A test that (ideally) executes a single well defined test objective (Testing Computer Software – Kaner, Faulk, Nguyen)
- A specific set of test data and associated procedures developed for a particular objectives (IEEE 729-1983)

# Why write test case?

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- Accountability
- Reproducibility
- Tracking
- Automation
- To find bugs
- To verify that tests are being executed correctly
- To measure test coverage

# Test case Essentials

- Tracking information
- Test case ID
- Test case description
  - Objective/Title
  - Steps
  - Test data: input/output/default
  - Expected results
  - Observed results
  - Status: Pass/Fail/Blocked/Skipped
- Test environment
- Script
- Bug ID
- Comments
- ...

# Test case Essentials

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- Test case Objective/Title
  - The most important essential
  - Gives reader a description and idea of the test
  - A good test name makes review easier
  - Easier to pass to another person, automation team
  - In many cases, may be the only part of the test case documented

# Test case Essentials

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- Test case Objective/Title Syntax
  - Action + Function + Operating Condition**
  - Action: Verify, Test, Validate, Execute, Run, Print...
  - Function: function, feature, validation point
  - Operating Condition: data, specific condition

| Action | Function      | Operating Condition                |
|--------|---------------|------------------------------------|
| Run    | annual report | from standard data (file location) |
| Run    | annual report | on Day 1 of fiscal year            |
| Run    | annual report | from empty spreadsheet             |
| Run    | annual report | on last day of fiscal year         |

# Test case Essentials

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- Validation Point
  - This is the expected result
  - Write it as a step
  - Define clearly state: what behavior, result or point that you are attempting to validate

# Test Case Template

| TC ID | Description                   | Steps                                                                                                                                 | Expected Result                                                                      | Observed result                                                                            | Status            |
|-------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------|
| TC001 | <i>/*Test<br/>objective*/</i> | <i>/*Very clear and<br/>specific steps*/</i><br><u>Pre-condition:</u><br><br><u>Steps:</u><br>1. Action 1<br>2. Action 2<br><br>..... | <i>/*you need to pre<br/>determine what<br/>your program is<br/>supposed to do*/</i> | <i>/* write down<br/>the result that<br/>you get when<br/>excecute the test<br/>case*/</i> | Passed/<br>Failed |

# What is a Good Test case?

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- Accurate – tests what it is designed to test
- Economical – no unnecessary steps
- Repeatable, reusable – keep going on
- Traceable – to a requirement
- Appropriate – for test environment
- Self standing – independent of the writer
- Self cleaning – picks up after itself

